Kamino DVI Solver Benchmark

RSL-RL · 300 iterations · three seeds · mean ± 95% Student-t CI

- Isaac-Ant-Direct: tuned DVI is 4.9× faster than current Kamino and 5.0× faster than PR3570 P-ADMM.
- Isaac-Ant-Direct: tuned DVI remains 2.5× slower than MJWarp and remains 1.4× slower than PhysX.
- Isaac-Cartpole-Direct: tuned DVI is 2.0× faster than current Kamino and 2.0× faster than PR3570 P-ADMM.
- Isaac-Cartpole-Direct: tuned DVI remains 1.5× slower than MJWarp and is approximately equal to PhysX.
- Isaac-Fourbar-Pole-Swingup: tuned DVI is 1.9× faster than current Kamino and 1.9× faster than PR3570 P-ADMM.
- Isaac-Velocity-Flat-AnymalD: tuned DVI is 6.8× faster than current Kamino and 7.1× faster than PR3570 P-ADMM.
- Isaac-Velocity-Flat-AnymalD: tuned DVI remains 2.4× slower than MJWarp and remains 1.9× slower than PhysX.

Task	Variant	Envs	Iteration [s]	Total FPS	Reward	Episode length	Success
Isaac-Ant-Direct	Kamino current	4096	2.040 ± 0.028	64263 ± 890	7402.741 ± 1650.401	898.716 ± 1.169	1.000 ± 0.000
Isaac-Ant-Direct	PR3570 tuned DVI	4096	0.413 ± 0.005	317194 ± 3709	10785.473 ± 3745.488	845.261 ± 218.999	0.863 ± 0.539
Isaac-Ant-Direct	PR3570 P-ADMM	4096	2.048 ± 0.023	63990 ± 722	8547.088 ± 4094.777	897.738 ± 3.259	1.000 ± 0.000
Isaac-Ant-Direct	MJWarp	4096	0.167 ± 0.001	785924 ± 4857	8872.347 ± 33.530	876.501 ± 25.822	0.921 ± 0.075
Isaac-Ant-Direct	PhysX	4096	0.304 ± 0.007	433084 ± 9769	6588.875 ± 1341.016	845.561 ± 18.860	0.888 ± 0.032
Isaac-Cartpole-Direct	Kamino current	4096	0.261 ± 0.005	251502 ± 4447	4.942 ± 0.005	300.000 ± 0.000	1.000 ± 0.000
Isaac-Cartpole-Direct	PR3570 tuned DVI	4096	0.133 ± 0.001	494241 ± 5153	4.950 ± 0.014	300.000 ± 0.000	1.000 ± 0.000
Isaac-Cartpole-Direct	PR3570 P-ADMM	4096	0.260 ± 0.002	252178 ± 2015	4.946 ± 0.014	300.000 ± 0.000	1.000 ± 0.000
Isaac-Cartpole-Direct	MJWarp	4096	0.087 ± 0.001	749904 ± 9903	4.947 ± 0.009	299.955 ± 0.193	1.000 ± 0.000
Isaac-Cartpole-Direct	PhysX	4096	0.133 ± 0.004	494030 ± 12633	-5.542 ± 0.151	293.953 ± 5.572	0.970 ± 0.009
Isaac-Fourbar-Pole-Swingup	Kamino current	4096	1.260 ± 0.016	52151 ± 607	3.283 ± 0.703	300.000 ± 0.000	N/A
Isaac-Fourbar-Pole-Swingup	PR3570 tuned DVI	4096	0.647 ± 0.029	102082 ± 4570	3.469 ± 0.438	300.000 ± 0.000	N/A
Isaac-Fourbar-Pole-Swingup	PR3570 P-ADMM	4096	1.256 ± 0.049	52374 ± 1841	3.862 ± 0.302	300.000 ± 0.000	N/A
Isaac-Velocity-Flat-AnymalD	Kamino current	4096	6.057 ± 0.183	16307 ± 423	21.668 ± 0.335	988.969 ± 6.014	0.997 ± 0.002
Isaac-Velocity-Flat-AnymalD	PR3570 tuned DVI	4096	0.889 ± 0.004	110716 ± 450	21.683 ± 0.509	977.112 ± 13.521	0.995 ± 0.005
Isaac-Velocity-Flat-AnymalD	PR3570 P-ADMM	4096	6.317 ± 0.107	15606 ± 254	21.722 ± 0.597	985.220 ± 3.083	0.997 ± 0.007
Isaac-Velocity-Flat-AnymalD	MJWarp	4096	0.371 ± 0.014	265486 ± 10653	15.859 ± 15.941	978.694 ± 31.812	0.750 ± 1.069
Isaac-Velocity-Flat-AnymalD	PhysX	4096	0.479 ± 0.012	205414 ± 5189	19.574 ± 2.938	986.785 ± 3.059	0.995 ± 0.003

Data quality and failures

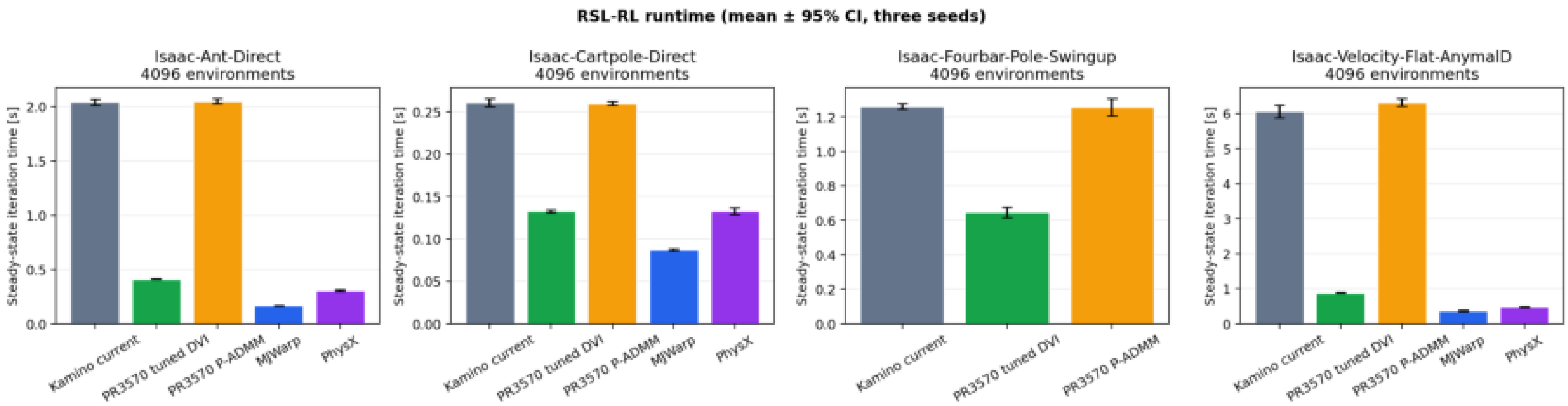
- Isaac-Ant-Direct: schema v1.1 success differs from TensorBoard in 15/15 runs and 3556/4500 points; report uses TensorBoard success.
- Isaac-Cartpole-Direct: schema v1.1 success differs from TensorBoard in 15/15 runs and 1600/4500 points; report uses TensorBoard success.
- Isaac-Fourbar-Pole-Swingup: learning.success_rate.series_per_iter and TensorBoard Metrics/success_rate are absent in 9/9 runs; this is a benchmark/task-stack bug. Report shows N/A.
- Isaac-Velocity-Flat-AnymalD: schema v1.1 success differs from TensorBoard in 0/15 runs and 0/4500 points; report uses TensorBoard success.
- IsaacContrib-DrLegs-Walk: schema v1.1 success differs from TensorBoard in 0/4 runs and 0/1200 points; report uses TensorBoard success.
- Isaac-Ant-Direct PR3570 tuned DVI has seed-sensitive weak learning: the three-seed success 95% CI half-width is 0.539; this is not a runtime or stability failure.
- Isaac-Velocity-Flat-AnymalD MJWarp has seed-sensitive weak learning: the three-seed success 95% CI half-width is 1.069; this is not a runtime or stability failure.
- Bundle workspace status: 43/58 completed full runs report versions.git_dirty=true; 58/58 expose a boolean status. This field includes untracked paths and is broader than the runner's tracked-source check; true does not by itself contradict exact-HEAD validation in current manifests.
- Legacy campaign source provenance: legacy bundles record 3 distinct commits. Runner manifests did not capture exact HEAD, and these runs did not pass the current clean-

Incomplete cells: descriptive successful seeds

Descriptive per-seed results from successful runs in incomplete cells. Excluded from summaries, plots, confidence intervals, and comparative speedups.

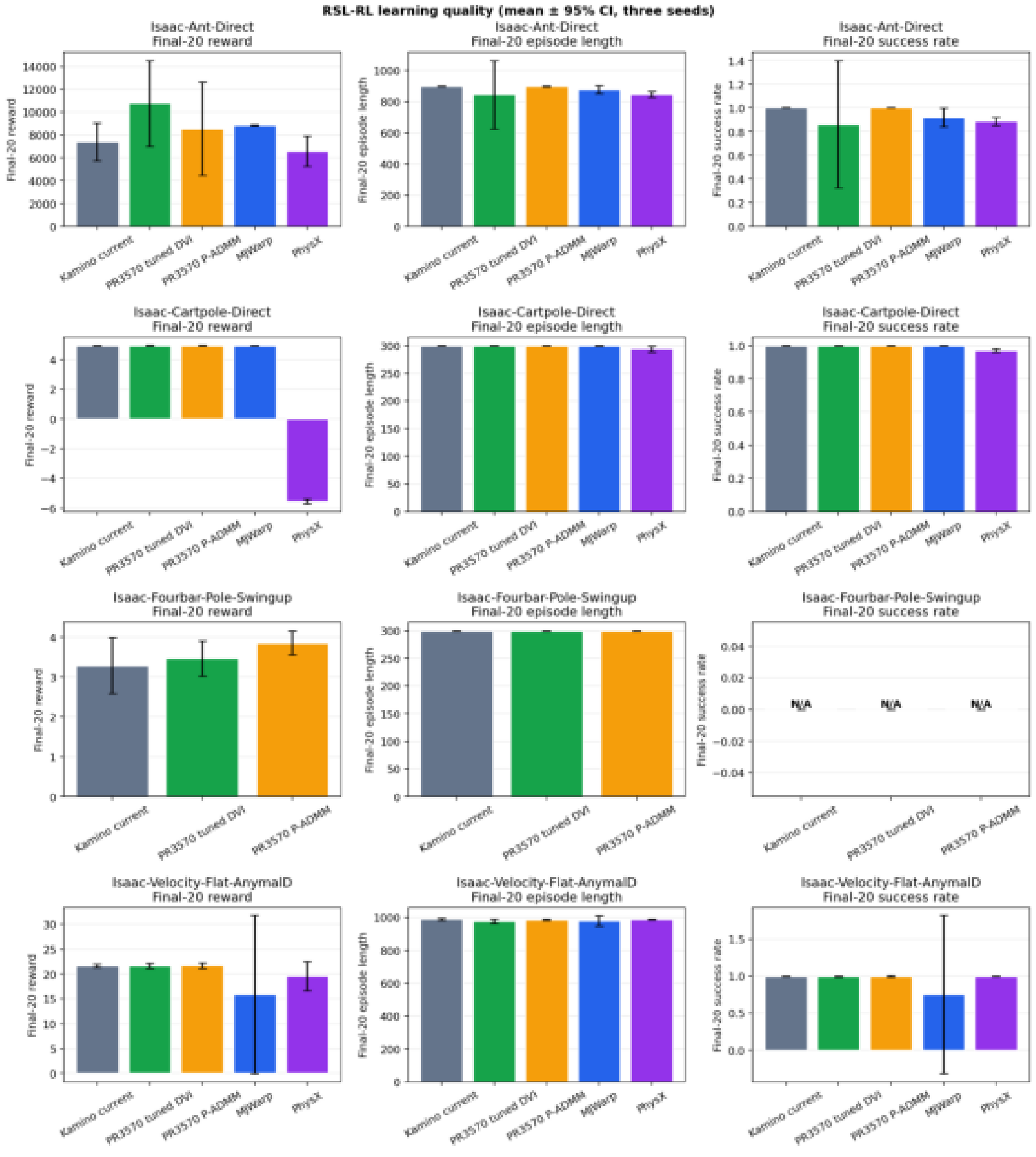
Task	Variant	Seed	Envs	Completed	Iteration [s]	Total FPS	Reward	Episode length	Success
IsaacContrib-DrLegs-Walk	Kamino current	42	4096	2/3	8.228	11981	247.393	469.982	0.000
IsaacContrib-DrLegs-Walk	Kamino current	44	4096	2/3	8.168	12082	48.200	467.342	0.000
IsaacContrib-DrLegs-Walk	PR3570 P-ADMM	42	4096	2/3	9.785	10047	7.735	24.799	0.065
IsaacContrib-DrLegs-Walk	PR3570 P-ADMM	44	4096	2/3	9.722	10112	7.690	24.483	0.065

Runtime



RSL-RL runtime (mean ± 95% CI, three seeds)
Isaac-Ant-Direct
4096 environments
Isaac-Cartpole-Direct
4096 environments
Isaac-Fourbar-Pole-Swingup
4096 environments
Isaac-Velocity-Flat-AnymalD
4096 environments
Steady-state iteration time [s]
Kamino current
PR3570 tuned DVI
PR3570 P-ADMM
MJWarp
PhysX

Learning


RSL-RL learning quality (mean ± 95% CI, three seeds)
Isaac-Ant-Direct
Final-20 reward
Final-20 episode length
Final-20 success rate
Isaac-Cartpole-Direct
Final-20 reward
Final-20 episode length
Final-20 success rate
Isaac-Fourbar-Pole-Swingup
Final-20 reward
Final-20 episode length
Final-20 success rate
Isaac-Velocity-Flat-AnymalD
Final-20 reward
Final-20 episode length
Final-20 success rate
Kamino current
PR3570 tuned DVI
PR3570 P-ADMM
MJWarp
PhysX
N/A